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Communication Technology

V2V, V2I, Network Architecture and Spectrum for Fog-Independent Haulage

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ABSTRACT

Communications carry the cooperative awareness layer, which is the highest-scoring single technology in the evaluation and the layer that works when every optical sensor has failed. This document specifies the message set, evaluates candidate bearers for the direct vehicle-to-vehicle link and the infrastructure backhaul, analyses the propagation environment of a ridge-and-pit mine, addresses congestion at fleet scale, sets out security requirements, covers the personal device link for pedestrian protection, and defines the RF survey that must precede bearer selection.

Prepared as part of a solution-design research package addressing dense-fog and low-visibility haul road operations in large-scale mechanised open cast iron ore mines. Figures relating to NMDC production, statutory provisions and commercial systems are compiled from public sources; site-specific values marked as indicative require validation against NMDC operational records before use in a formal proposal.

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1. The Role of Communications

In the proposed architecture, communications is not a supporting utility. It carries layer L3, the cooperative awareness layer, which scored highest of any single technology in the Volume II Document 2 evaluation and which has three properties no sensor can match:

- ? **Complete immunity to fog.** RF at the frequencies concerned is unaffected by water droplets by many orders of magnitude.
- ? **Non-line-of-sight capability.** A radar cannot see around a switchback; a radio can be heard around one. On a hairpin ramp, two trucks 80 m apart along the road may be separated by a rock face. The cooperative layer is the only technology in the architecture that addresses this geometry, and the geometry is common at Bailadila.
- ? **Identity and intent, not merely presence.** A radar return says something is there. A cooperative message says *which* truck it is, how fast, in which direction, loaded or empty, braking or accelerating, reversing or tipping. Intent is what allows the threat assessment of FR-19 to distinguish a normal interaction from a developing conflict - and therefore what allows nuisance alerts to be suppressed without suppressing real ones.

The sector precedent is strong. The SAFEmine lineage within Hexagon's mining CAS is described as providing 360 degree protection at any speed and any visibility with a typical range of 800 m, without requiring line of sight, and the resulting system family is installed on more than 65,000 units across over 100 mine sites. The architecture pattern is proven at scale; what this document addresses is bearer selection for the Bailadila environment.

Equally important is what communications must not do. Per principle P4 and requirement CR-06, no safety function may depend on infrastructure. The direct vehicle-to-vehicle link must function with every base station, access point and backhaul link down. Infrastructure improves coordination; it never provides protection.

2. The Message Set

2.1 Basic Status Message (BSM)

Broadcast by every equipped vehicle at not less than 10 Hz (FR-18, PR-14).

Field	Content	Purpose
Identity	Persistent vehicle identifier	Association, tracking, control room picture
Class	Haul truck, dozer, grader, water tanker, shovel, light vehicle, personnel carrier	Threat weighting; a light vehicle near a truck is a different case from two trucks
Position	Latitude, longitude, height in mine datum	Core
Position confidence	Horizontal protection level from L1	Receiver must know how much to trust the sender's position
Velocity	Full vector	Closing rate, trajectory prediction
Heading	From dual-antenna GNSS	Valid at zero speed, unlike course over ground
Acceleration	Longitudinal and lateral	Short-horizon prediction; braking detection
Dimensions	Length, width, height, and reference point offset	Envelope-accurate conflict computation; a 100 t truck is not a point
Operating state	Loaded, empty, stationary, reversing, tipping, loading, broken down	Intent; context suppression of nuisance alerts
Brake and retarder state	Applied or not	Immediate intent signal

Field	Content	Purpose
Road segment and lane	From the local road model	Rapid association; disambiguates switchback geometry
System health	Nominal, degraded, failed; effective sensing range	Receiver knows whether the sender can see non-cooperative hazards
Timestamp	Synchronised time of validity	Latency compensation; sequencing

Two fields distinguish this message set from a conventional automotive BSM, and both matter here. **Position confidence** allows a receiver to widen its conflict envelope when a sender's position is uncertain - without it, a truck with a degraded GNSS fix appears as precisely located as one with a clean RTK solution. **System health and effective sensing range** tell a receiver whether the sender is able to detect non-cooperative hazards, which supports both the control room picture and a more subtle behaviour: a truck following one whose radar is blocked should not assume the leader will react to an obstruction.

2.2 Other message types

Message	Rate	Purpose
Personal Safety Message	1-5 Hz	Position and identity of equipped personnel; FR-21
Hazard Marker	Event-driven, then periodic	Washout, rockfall, road maintenance, stalled machine, blast exclusion; FR-22
Road Model Update	On change	Distribution of updated road geometry; FR-07
Movement Authority	On grant and revoke	Slot allocation and permission on critical sections at V3-V4; OR-04
Visibility Class Update	On change, periodic refresh	Per-zone operating class from L0; FR-01, FR-02
Emergency	Immediate, repeated	Emergency stop, incident declaration, evacuation
Health and Telemetry	Low rate	Control room fleet health; OR-07

Hazard markers deserve particular attention at Bailadila. Volume III Document 1, Section 7 established that radar cannot reliably detect a washout - smooth water reflects specularly away from the sensor. Washouts are a monsoon hazard, occurring in exactly the conditions where nothing else can see them either. The hazard marker message, placed by a road maintenance crew or the control room and broadcast to approaching vehicles, is the **only mechanism in the entire architecture that addresses this hazard**. Its reliability and the procedures around its use are correspondingly important.

3. V2V Bearer Evaluation

3.1 Candidate technologies

Technology	Band	Latency	Range	Density handling	Maturity	Mining use
C-V2X PC5 (LTE-V2X, NR-V2X)	5.9 GHz ITS	1-20 ms	300-1500 m	Good; semi-persistent scheduling	High, growing	Emerging
ITS-G5 / DSRC (802.11p)	5.9 GHz ITS	1-10 ms	300-1000 m	Moderate; CSMA contention degrades under load	High	Some
Proprietary sub-GHz CAS	400-900 MHz	10-50 ms	~800 m, NLOS tolerant	Good	High in mining	Established

Technology	Band	Latency	Range	Density handling	Maturity	Mining use
UWB	3.1-10.6 GHz	Low	50-200 m	Good at short range	High	Established for tagging
Wi-Fi mesh (802.11 a/n/ac)	2.4/5 GHz	10-50 ms, variable	100-300 m	Moderate	High	Common
Private LTE/5G Uu	Licensed/shared	20-100 ms	Cell dependent	Excellent	High	Growing

3.2 C-V2X PC5 direct

Principle. Direct device-to-device communication over the PC5 sidelink, operating without any network infrastructure. Mode 4 (LTE-V2X) and Mode 2 (NR-V2X) provide autonomous resource selection: each device senses the channel and selects transmission resources semi-persistently, so access is scheduled rather than contended.

Strengths for this application. Infrastructure-independent by design, satisfying CR-06 directly. Semi-persistent scheduling degrades more gracefully than CSMA contention as density rises, which matters at the 60-vehicle density of CR-03. Strong range, supported by an automotive ecosystem driving cost and ruggedness. A clear evolution path from LTE-V2X to NR-V2X with higher reliability and lower latency, and NR-V2X adds unicast and groupcast with feedback, enabling acknowledged delivery for critical messages such as movement authority.

Weaknesses. Less mining field history than the incumbent proprietary bearers. Operates at 5.9 GHz, where diffraction around terrain is poorer than at sub-GHz - a material consideration in a ridge and pit environment (Section 4). Indian spectrum position for the ITS band in a private industrial application requires confirmation.

3.3 ITS-G5 / DSRC

Principle. IEEE 802.11p, a Wi-Fi derivative adapted for vehicular use, with CSMA/CA channel access.

Strengths. Very low latency; mature; simple; extensive field deployment in intelligent transport systems; no infrastructure required.

Weaknesses. CSMA contention degrades under high load, with hidden-node problems in obstructed terrain - and a pit is a hidden-node environment by construction: two trucks that cannot hear each other because a rock face is between them will transmit simultaneously, and a third that hears both receives a collision. This is a specific and serious concern at Bailadila. Same 5.9 GHz propagation limitation as C-V2X.

3.4 Proprietary sub-GHz CAS bearers

Principle. Dedicated narrowband or spread-spectrum radio in sub-GHz bands, purpose-designed for mining collision avoidance, with an optimised protocol for periodic position broadcast among a bounded vehicle population.

Strengths. Purpose-built and proven in exactly this application, with the documented non-line-of-sight capability that matters most at Bailadila. Sub-GHz propagation diffracts around terrain and penetrates obstructions far better than 5.9 GHz - the single most relevant technical property for a ridge-and-switchback environment. Long range at modest power. Large deployed base in mining.

Weaknesses. Proprietary, with vendor lock-in risk that conflicts with requirement IR-01. Lower data rate constrains message richness - the extended message set of Section 2 may not fit. Smaller ecosystem and slower cost reduction than the automotive-driven alternatives. Spectrum authorisation in India requires confirmation for the specific band used.

3.5 Comparative assessment and recommendation

Criterion	C-V2X PC5	ITS-G5	Sub-GHz CAS
Latency (PR-13, <=100 ms)	Excellent	Excellent	Good
Update rate (PR-14, >=10 Hz)	Excellent	Excellent	Good
Range (CR-02, >=500 m LOS)	Excellent	Good	Excellent
NLOS in ridge/pit terrain	Moderate	Moderate	Excellent
Density (CR-03, 60 vehicles)	Excellent	Moderate	Good
Infrastructure independence (CR-06)	Excellent	Excellent	Excellent
Message richness	Excellent	Excellent	Limited
Security framework	Mature (ITS PKI)	Mature (ITS PKI)	Vendor-specific
Openness (IR-01)	Open standard	Open standard	Proprietary
Mining field proof	Limited	Limited	Extensive
Cost trajectory	Falling (automotive volume)	Falling	Flat

Recommendation. C-V2X PC5 direct is preferred on the merits - open standard, best density handling, richest message set, mature security framework, and a falling cost curve driven by automotive volume.

However, the recommendation is conditional. The one criterion on which the proprietary sub-GHz option is clearly superior - non-line-of-sight propagation in ridge and pit terrain - is also the criterion most specific to Bailadila and most likely to determine real performance. A bearer with ideal specifications that cannot reach a truck around a switchback is worse than a bearer with modest specifications that can.

The bearer decision must be resolved by on-site RF propagation measurement, not by specification comparison. Section 12 specifies the survey. If the survey shows acceptable 5.9 GHz coverage across the haul road network, C-V2X PC5 should be selected. If it shows material coverage holes at switchbacks, ramp transitions and pit-bottom sections, a dual-bearer architecture - sub-GHz for a compact critical message subset, C-V2X for the full message set where coverage permits - should be adopted, accepting the additional complexity in exchange for coverage where it matters most.

4. Propagation in a Ridge and Pit Environment

Bailadila presents one of the more difficult RF environments in surface mining.

Feature	Effect
Ridge topography with 400-700 m relief	Terrain shadowing between benches and between complexes
Deep pits with high benched walls	Severe shadowing; strong reflections from rock faces
Switchback ramps	Adjacent road levels separated by rock; the classic NLOS case
High-grade haematite	Highly conductive; strong reflection and multipath
Large steel vehicle bodies	Mobile scatterers and mobile obstructions; a truck can shadow another
Dense vegetation on undisturbed slopes	Attenuation, seasonally variable with monsoon foliage
Heavy rain	Negligible below 6 GHz; a genuine consideration only at millimetre-wave

Frequency and diffraction. Diffraction loss around an obstruction falls as frequency falls. A sub-GHz signal bends around a bench edge or rock outcrop substantially better than a 5.9 GHz signal. For a ridge-and-switchback mine this is the dominant propagation consideration, and it is the physical basis of the caution in Section 3.5.

Mitigations available regardless of bearer.

- ? **Relay through intermediate vehicles.** A truck that hears both parties can relay, extending effective coverage without infrastructure. This adds latency and complexity but exploits fleet density, which is highest in exactly the congested areas where coverage is worst.
- ? **Strategically sited repeaters.** A small number of repeaters at ramp heads and ridge crests can transform coverage. These are the one class of fixed infrastructure that is clearly justified, though they must satisfy the relocatability and solar autonomy requirements of ER-08.
- ? **Antenna placement.** Roof-mounted, well-clear antennas on a haul truck are substantially better than body-mounted. Diversity antennas mitigate multipath fading.
- ? **Coverage-aware operating procedures.** Where the survey identifies a persistent coverage hole, the control room can manage that section by movement authority at V3-V4 rather than relying on peer-to-peer awareness - an operational mitigation for a physical limitation.

5. Congestion and Scalability

Requirement CR-03 sets a design density of 60 vehicles within mutual range. At 10 Hz per vehicle this is 600 messages per second on a shared channel, before personal devices, hazard markers and other message types.

Mitigations.

- ? **Adaptive rate control.** Reduce broadcast rate for stationary vehicles and for those distant from any other vehicle; maintain full rate for those in motion near others. Congestion control of this kind is standard in ITS stacks and can reduce offered load substantially without reducing safety-relevant information.
- ? **Message prioritisation.** Emergency and movement authority messages pre-empt routine status broadcasts.
- ? **Compact encoding.** Efficient binary encoding rather than verbose formats.
- ? **Semi-persistent scheduling** (inherent to C-V2X PC5), which avoids the contention collapse that afflicts CSMA at high load.
- ? **Spatial reuse.** Vehicles far apart may reuse resources; the terrain shadowing that harms coverage helps here.
- ? **Channel separation.** Where spectrum permits, separate routine status broadcast from bulk data such as road model updates, so that a model push cannot delay a safety message.

Validation. Requirement V5 of the radar validation plan has a communications analogue: congestion behaviour must be tested at representative density, with loaded channel measurements rather than analytical estimates. Congestion failures are non-linear - performance is adequate until a threshold, then degrades sharply - so testing must probe beyond the design density, not up to it.

6. V2I and Backhaul Architecture

Function. Road model distribution, hazard marker propagation, movement authority, visibility class distribution, dispatch integration, telemetry and health upload, event log upload, software and model update.

None of this is safety-critical in the sense of CR-06. Loss of backhaul degrades coordination and control room visibility; it must not degrade protection.

Bearer	Assessment
Private LTE / 5G	Recommended. Good coverage from few sites, manageable in difficult terrain, strong security, supports both vehicle and fixed-node connectivity, established in mining
Wi-Fi mesh	Common in mining; weaker on roaming, coverage per node and latency consistency. Viable where already deployed
Existing site network	Reuse per CR-08 wherever coverage permits
LPWAN (LoRaWAN, NB-IoT)	Suitable for low-rate telemetry from solar roadside nodes; unsuitable for road model distribution

Store-and-forward is mandatory (CR-05). Given the terrain, every vehicle will spend time out of backhaul coverage. Design implications: vehicles cache the road model locally and operate from cache; events are buffered and uploaded opportunistically; road model updates propagate on next contact and may also **propagate vehicle-to-vehicle**, so that a truck that has received an update can pass it to one that has not - an epidemic distribution pattern well suited to a fleet that mixes continuously and a network with holes.

7. Personal Devices and Pedestrian Protection

Requirement FR-21 calls for personnel to carry a compatible device providing cooperative presence.

Requirement	Specification
Battery life	Full shift plus margin; target 16 hours; unambiguous low-battery indication
Form factor	Integrated into cap lamp, belt or vest; must not require a separate item to remember
Position source	GNSS where available; RF ranging to vehicles where not
Broadcast rate	1-5 Hz; higher when a vehicle is near
Alert to wearer	Vibration and audible; must penetrate PPE and ambient noise
Robustness	IP67, drop-tested, operable with gloves
Fault indication	Unmistakable indication of failure or flat battery

Layered protection. The personal device is one of three pedestrian protections, and the layering is deliberate:

- ? **Cooperative layer** - the personal device, giving long-range advance awareness to approaching vehicles.
- ? **Radar with micro-Doppler classification** - non-cooperative detection of anyone without a device, at the PR-03 range of 25 m.
- ? **Magnetic-field PDS** - short-range, non-line-of-sight, covering the immediate blind zone including positions radar cannot illuminate.

No single layer is sufficient. The cooperative layer fails for anyone without a working device - a contractor, a visitor, someone whose battery is flat. Radar fails for a person directly behind structure. Magnetic PDS covers only metres. Together they are adequate; individually none is.

The honest caveat: a person on foot on a haul road in dense fog remains at serious risk regardless of technology, because their own situational awareness is as degraded as everyone else's. **The primary control remains keeping people off haul roads during low-visibility operations**, with technology as the secondary control for when that fails. The system should never be presented as making pedestrian presence on a fog-bound haul road acceptable.

8. Security

Requirement CR-04 demands message authentication, and requirement IR-05 demands secured equipment. The threat is not primarily malicious in this environment - it is spoofing, replay and equipment tampering, with the additional consideration of the region's security context (constraint C-24).

Threat	Consequence	Control
Position spoofing	False vehicle injected; nuisance alerts, or masking of a real vehicle	Message signing with per-vehicle credentials
Replay	Stale position replayed as current	Timestamps within a validity window; sequence numbers
Identity spoofing	Impersonation of another vehicle	Certificate-based identity
Message injection	False hazard marker or movement authority	Authorisation - only credentialed sources may issue these message types
Jamming	Cooperative layer denied	Radar and map layers continue (principle P3); detect and report jamming; control room alerted
Equipment tampering or removal	Vehicle becomes invisible to the cooperative layer	Tamper detection and reporting per IR-06; heartbeat monitoring per OR-07
Firmware compromise	Arbitrary behaviour	Signed firmware; authenticated update only per IR-05

Practical approach. A site-operated PKI issuing per-vehicle credentials, with message signing on the safety-relevant message set. The computational cost of signing at 10 Hz is manageable on automotive hardware with cryptographic acceleration, and the ITS security frameworks developed for C-V2X and ITS-G5 provide a mature model to adopt rather than invent.

Privacy. The same infrastructure that authenticates also identifies, and identification of vehicles means identification of operators (constraint C-19, requirement IR-04). The data governance framework must be settled before deployment. Security design and privacy design are the same conversation and should be held once, early, with operator representation.

9. Time Synchronisation

Requirement IR-03 demands common time across all vehicles and infrastructure. This is more consequential than it appears: cooperative position reports are timestamped, and a receiver must compensate for transmission and processing delay to compare a remote vehicle's position at time T with its own at time T. A synchronisation error translates directly into a position error - at 10 m/s, 10 ms of error is 0.1 m, and 100 ms is 1 m, which is the lane margin.

Source. GNSS provides precise time to every equipped vehicle inherently, at no additional cost, with accuracy far exceeding the requirement. During GNSS outage, the local oscillator holds over; drift over the outage durations concerned is negligible. Fixed infrastructure synchronises by GNSS or by network time protocol.

Requirement. Better than 1 ms across the fleet, readily achievable by GNSS, which corresponds to a position error of 1 cm at haul speeds.

10. Spectrum and Regulatory Position

This is a long-lead item and risk R9 in the programme register.

Item	Position	Action
76-81 GHz radar	Automotive allocation; use in a private mining application requires confirmation	Confirm with WPC, Ministry of Communications, in Phase 0
5.9 GHz ITS band for C-V2X	Indian allocation and licensing terms for private industrial use require confirmation	Confirm early; this determines whether the preferred bearer is available
Sub-GHz proprietary CAS bands	Vendor-specific; some operate in licence-exempt bands, others require authorisation	Confirm per candidate vendor
UWB	Indian regulatory position for the intended bands	Confirm if UWB ranging is adopted
Private LTE / 5G	Spectrum options for captive networks in India have evolved; confirm current position	Confirm in Phase 0
Equipment type approval	All radio equipment requires applicable Indian type approval	Build into procurement; verify vendor holds approval, not merely that approval is possible

Coexistence. Requirement ER-09 and CR-07 demand demonstrated non-interference with existing site systems, with **explicit verification against blast initiation systems**. This is a safety-critical verification, not a formality, and must be conducted with the site's blasting authority.

11. Network Architecture Summary

Tier	Bearer	Function	Infrastructure dependence	Safety criticality
Tier 1 - Direct V2V	C-V2X PC5 preferred, subject to RF survey; sub-GHz CAS alternative or complement	BSM, Personal Safety Message, Hazard Marker, Emergency	None	Safety-critical
Tier 2 - V2I	Private LTE/5G or existing site network	Road model, movement authority, visibility class, dispatch, telemetry, updates	Full	Coordination only
Tier 3 - Roadside telemetry	LPWAN or LTE	Visibility and met node reporting	Gateway	Operational
Tier 4 - RTK corrections	Site RF network, with fallback	GNSS corrections to vehicles	Base and network	Degrades positioning to DGPS or standalone; guidance withdrawn if protection level exceeds lane margin
Tier 5 - Fallback	VHF/UHF voice	Human coordination when systems are unavailable	Repeaters	Retained as the ultimate fallback

Tier 5 should be retained indefinitely and exercised periodically. Voice radio is the control that currently permits limited movement in fog. A new system should supersede it in normal use and never remove it, because the one certainty about any deployed system is that it will at some point be unavailable, and the operators must retain the procedural competence to work without it.

12. RF Survey Specification

The bearer decision rests on this survey. It should be conducted in Phase 0, before procurement.

Item	Specification
Objective	Characterise propagation across the full haul road network of both complexes at candidate frequencies, and determine whether 5.9 GHz coverage is adequate or a sub-GHz bearer is required
Frequencies	5.9 GHz (C-V2X / ITS-G5 candidate); a representative sub-GHz frequency in the 400-900 MHz range; 2.4/5 GHz if Wi-Fi is under consideration; private LTE/5G candidate bands
Method	Vehicle-mounted transmitter and receiver at representative antenna heights for a haul truck and a light vehicle; continuous received-signal-strength logging with precise position; both directions on every haul road segment
Geometries	Straight ramp sections; switchbacks with both vehicles on adjacent levels; crest and sag curves; pit bottom to ramp head; bench to bench; complex to complex; congested areas at the loading face and crusher tip
Conditions	Dry season and monsoon, to capture the effect of wet vegetation and wet rock surfaces on propagation
Obstructions	Explicitly test with a loaded haul truck between transmitter and receiver, to quantify vehicle shadowing
Outputs	Coverage maps per frequency; identification of coverage holes with position and extent; recommended repeater siting; empirical link budget; bearer recommendation
Concurrent activity	Conduct the GNSS availability survey (Volume III Document 2, Section 1.4) with the same vehicle on the same runs. Both require driving the entire network with a logging vehicle; combining them roughly halves the cost and produces a single consistent picture of where the difficult sections are

The concurrency recommendation in the last row is a small point with disproportionate value. The GNSS survey, the RF survey and the instrumented radar campaign (Volume III Document 1, V3) all require a single logging vehicle driving the haul road network over a period spanning a monsoon. Combining them into one Phase 0 field campaign - one vehicle, one instrumentation fit, one dataset - produces the positioning, communications and radar evidence base simultaneously and at a fraction of the cost of three separate exercises. This single campaign is, in the assessment of this research package, the highest-value action available to the programme and should be initiated at the earliest monsoon window.